Geometric Attractor Convergence in a Viable System Kernel

Executive Summary

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What We Did

We built a computational kernel (`vsf-nucleus`) based on Beer's Viable System Model (VSM) and tested whether a system governed by a small set of cybernetic axioms converges to a stable attractor in state space. The kernel's state is a point in $\mathbb{R}^{128}$, structured into semantic blocks: identity, resources, entropy, operations, communication channels, attractor memory, and regulatory intelligence.

We ran twelve experiments on physical hardware (Intel Core i7-7600U, Proxmox hypervisor) from April 2026 onward, measuring convergence, basin geometry, thermal resilience, observer effects, architecture ablation, and — in the final experiment — the whole assembled system with every VSM layer running live.

Headline Results

1. The attractor is a stable point sink, confirmed three independent ways. We measured the largest Lyapunov exponent directly (Benettin's method): $\lambda_{\max} = -0.010050 = \ln(0.99)$ per tick, to machine precision — negative, so nearby trajectories contract rather than diverge. A Monte-Carlo estimate puts the basin volume at ≈ 1.0 over the reachable region, and persistent-homology topology gives $\beta_0 = 1,\ \beta_1 = 0$ (one component, no hole, no limit cycle). Dynamics, volume, and shape agree: a single stable point.

2. A coordinate-free viability certificate, $V = \min(g, c, t, m)$. We distil the measurements into one functional built from intrinsic invariants (feasibility, contraction sign, topological signature, noise margin). On this kernel the binding axis is $g = 0.102$ (the CPU-budget constraint), and breaking any single axis independently drives $V \to 0$ — so V is a discriminating vector, not a cosmetic scalar that is always near one.

3. The observer is constitutive, not passive. Structured observation measurably lowers the system's entropy by 0.31 bits and compresses its effective dimension from ≈ 5.0 to 3.9 — a measured instance of second-order cybernetics, where the observer is part of the dynamics it stabilises.

Key Finding: Two-Layer Viability

The ablation experiments reveal a **two-layer viability structure**:

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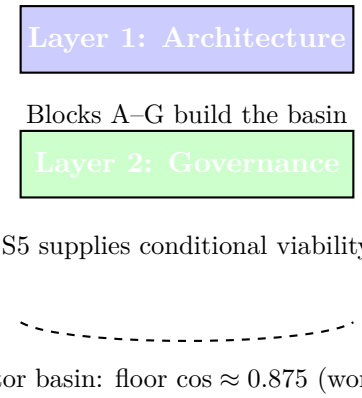


Figure 1: Architecture builds the basin and its homeostasis; governance is the difference between survival and collapse under stress.

Layer 1: Architecture The arrangement of VSM blocks physically determines which regions of state space are habitable. The bare substrate (no blocks) drifts off-target and never stabilises; the blocks are what create the $\mathbb{R}^{128}$ point sink.

Layer 2: Governance (S5) At rest, governance is *redundant* — the system converges identically with or without it. Under stress it is decisive. Governance has two arms: a *reactive* one (the algedonic safety response that throttles a runaway thermal load) and a *deliberative* one (the constitutional covenant veto that can halt the system's own autonomy). The experiments exercise both.

When both layers are present the system holds viability at cos ≈ 0.875 (on a 0–1 scale, 0.70 = viable threshold) even under maximum impulsive stress — a margin of 0.175 never breached in any experiment.

The Capstone: an Operationally Closed System

The final experiment ran the complete kernel with all five of Beer's systems (S1–S5) live, under a single perturbation. Each level engaged within its own cybernetic scope. The decisive observation traverses the whole model in one tick: operations (S1) registered a dip, coordination (S2) delivered a policy signal, control (S3) detected the degradation and *attempted* a correction, and policy (S5) *overruled* it — after which the system self-recovered. A separate test confirmed the constitutional veto end-to-end: it halts autonomous control under sustained degradation and restores it only on an explicit, cryptographically recorded human resume.

This is the point of the architecture: not a simulation of Beer's diagram, but a system in which each arrow carries a measured event.

Empirical Results

Property	Measure	Status
Lyapunov exponent	$\lambda_{\max} = \ln(0.99) < 0$	Measured, point sink
Basin volume	≈ 1.0 (reachable region)	Monte-Carlo
Topology	$\beta_0 = 1,\ \beta_1 = 0$	No limit cycle
Basin uniqueness	$\cos_{\text{mean pairwise}} = 0.9854$	Confirmed
Thermal viability	viable at 68 °C	Holds
Basin floor (impulsive)	$\cos \approx 0.875$	Margin 0.175
Observer entropy drop	$\Delta H = 0.31$ bits	Constitutive
Viability certificate	$V = \min(g, c, t, m),\ g = 0.102$	Discriminates
Real-process governance	4 OS processes, $\tau = 1.0$	cgroup v2 live
VSM operational closure	S1–S5 causal in one run	Demonstrated

What This Means

1. **Viability is geometric.** The system's stability emerges from the geometry of its state space, not from external enforcement. This is what Beer meant by a *viable system*.
2. **Ethical constraints enter as feasibility conditions.** The axioms that prohibit harm and coercion are not bolted on — a covenant violation collapses viability ($V \to 0$) exactly as exhausting the CPU budget does. The mathematics is *indifferent* to which constraints encode ethics; that indifference is precisely what makes it a general certifier rather than a moral theory.
3. **Governance is conditional resilience.** A well-designed policy layer does not *create* viability — the architecture does that — but it is the difference between graceful survival and collapse when the system is challenged.
4. **The basin is deep.** Under every tested perturbation the system never approached the viability threshold (0.70); the nearest approach was 0.875, a 12.5% margin above failure.

Honest Limits

The work runs on a single hardware node; the thermal stressor, while it drives real silicon temperature, is a synthetic workload; and most runs are closed-system. The final, sharpest limit is informative: because the attractor holds its viability measure invariant even against a CPU load more than twice the core count, the system's *anticipatory* channel cannot read an early warning from its own regulated state. Genuine anticipation requires an external environment — variables whose variance is independent of the kernel's own attractor. Coupling the kernel to such an environment is the principal direction of ongoing work.

Next Steps

1. **External environment coupling.** Connect the kernel to genuinely external operational units (the motivating case: a distributed climate-sensing network) whose signals are not slaved to the kernel's attractor — the test the anticipatory channel actually needs.
2. **Inter-system coordination via a formal substrate.** A companion line of work studies whether two heterogeneous models can coordinate as distinct VSM layers through a shared formal language (the basis of a second paper).

3. **vOS on bare metal.** Deploy the kernel against real workloads (databases, inference) to measure attractor geometry under diverse load.

For the Beer Community

This work returns to first principles: *what does it mean for a system to be viable?* We give a computational answer grounded in cybernetic axioms, with instruments — Benettin's algorithm, persistent homology, cryptographic attestation — that did not exist when Beer wrote. The architecture–governance distinction, and the demonstration that the five systems close an operational loop on real hardware, suggest the blocks of the VSM are not metaphors: they are viability requirements with measurable geometry.

The code is open source. The complete paper, data, and experimental logs are available on request.